

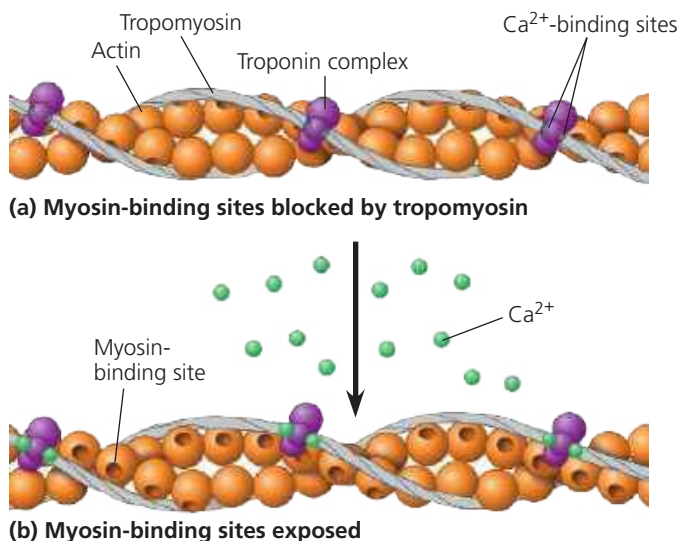
At rest, most muscle fibers contain only enough ATP for a few contractions. Powering repetitive contractions requires two other storage compounds: creatine phosphate and glycogen. The transfer of a phosphate group from creatine phosphate to ADP in an enzyme-catalyzed reaction synthesizes additional ATP. In this way, the resting supply of creatine phosphate can sustain contractions for about 15 seconds. ATP stores are also replenished when glycogen is broken down to glucose. During light or moderate muscle activity, this glucose is metabolized by aerobic respiration. This highly efficient metabolic process yields enough power to sustain contractions for nearly an hour. During intense muscle activity, oxygen becomes limiting and ATP is instead generated by lactic acid fermentation (see Concept 9.5). This anaerobic pathway, although very rapid, generates much less ATP per glucose molecule and can sustain contraction for only about 1 minute.

### The Role of Calcium and Regulatory Proteins

Proteins bound to actin play crucial roles in controlling muscle contraction. In a muscle fiber at rest, **tropomyosin**, a regulatory protein, and the **troponin complex**, a set of additional regulatory proteins, are bound to the actin strands of thin filaments. Tropomyosin covers the myosin-binding sites along the thin filament, preventing actin and myosin from interacting (**Figure 50.29a**).

Motor neurons enable actin and myosin to interact by triggering a release of calcium ions ( $\text{Ca}^{2+}$ ) into the cytosol. Once in the cytosol,  $\text{Ca}^{2+}$  binds to the troponin complex, causing the myosin-binding sites on actin to be exposed (**Figure 50.29b**). Note that the effect of  $\text{Ca}^{2+}$

▼ **Figure 50.29 The role of regulatory proteins and calcium in muscle fiber contraction.** Each thin filament consists of two strands of actin and associated regulatory proteins: two long molecules of tropomyosin and multiple copies of the troponin complex.



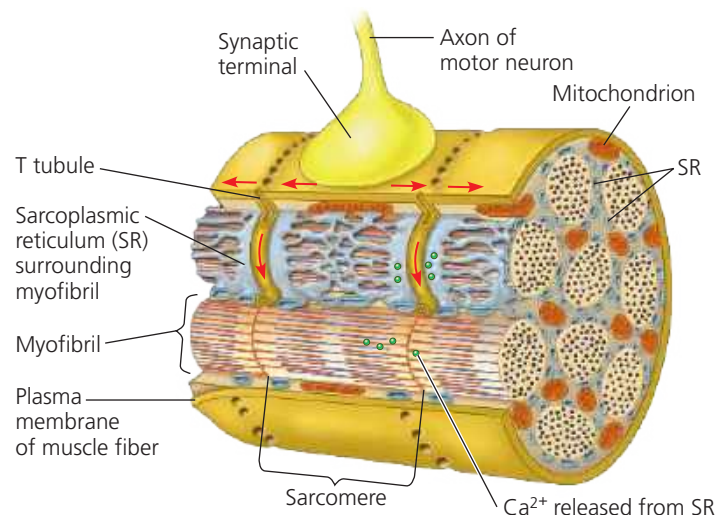
is indirect: Binding to  $\text{Ca}^{2+}$  causes the troponin complex to change shape, dislodging tropomyosin from the myosin-binding sites.

When the  $\text{Ca}^{2+}$  concentration rises in the cytosol, the cycle of cross-bridge formation begins, the thin and thick filaments slide past each, and the muscle fiber contracts. When the  $\text{Ca}^{2+}$  concentration falls, the binding sites are covered, and contraction stops.

Motor neurons cause muscle contraction through a multi-step process triggering the movement of  $\text{Ca}^{2+}$  into the cytosol of muscle cells. This process is summarized in **Figure 50.30** and detailed step by step in **Figure 50.31**. First, the arrival of an action potential at the synaptic terminal of a motor neuron ❶ causes release of the neurotransmitter acetylcholine. Binding of acetylcholine to receptors on the muscle fiber leads to a depolarization that initiates an action potential. Within the muscle fiber, the action potential spreads deep into the interior, following infoldings of the plasma membrane called **transverse (T) tubules**. ❷ These make close contact with the **sarcoplasmic reticulum (SR)**, a specialized endoplasmic reticulum. As the action potential spreads along the T tubules, it triggers changes in the SR, opening  $\text{Ca}^{2+}$  channels ❸. Calcium ions stored in the interior of the SR flow through open channels into the cytosol ❹ and bind to the troponin complex, ❺ initiating the muscle fiber contraction.

When motor neuron input stops, the filaments slide back to their starting position as the muscle relaxes. Relaxation

▼ **Figure 50.30 The roles of the sarcoplasmic reticulum and T tubules in muscle fiber contraction.** The synaptic terminal of a motor neuron releases acetylcholine, which depolarizes the plasma membrane of the muscle fiber. The depolarization causes action potentials (red arrows) to sweep across the muscle fiber and deep into it along the transverse (T) tubules. The action potentials trigger the release of calcium (green dots) from the sarcoplasmic reticulum into the cytosol. Calcium initiates the sliding of filaments by allowing myosin to bind to actin.



➡ Mastering Biology BioFlix® Animation: Muscle Structure